FINKRAFT B2B E-INVOICING PLATFORM

UAE Federal Tax Authority (FTA) Peppol 5-Corner DCTCE Compliance Platform

CTO Take-Home Assignment Deliverable | Architecture & Editable State Machine

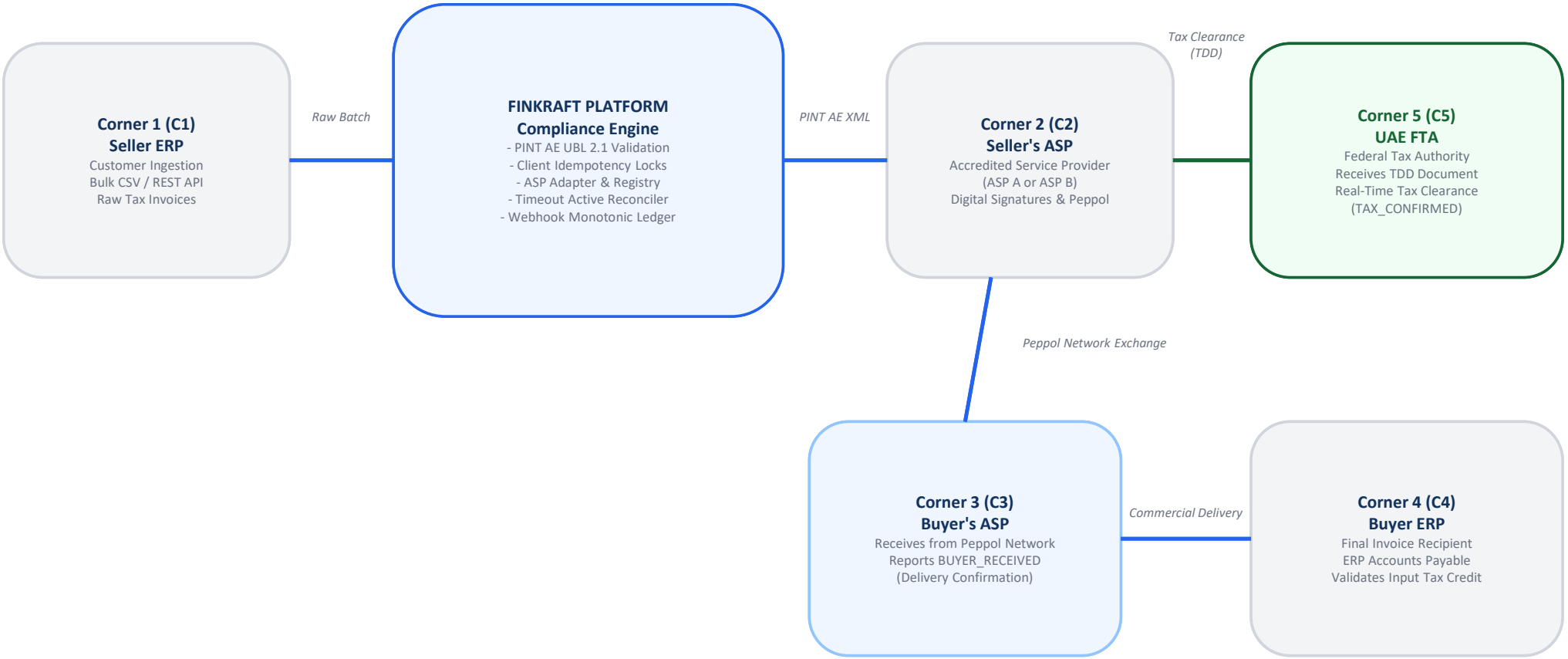
Part 1: Architecture Design Note
Unified Adapter & State Lattice

Part 2: Working Python Codebase
24 Tests, Batch Processor, Zero Deps

Part 3: 30-Day CTO Execution Plan
6-Week Roadmap for 3 BE + 1 FE

UAE FTA 5-Corner DCTCE Architecture

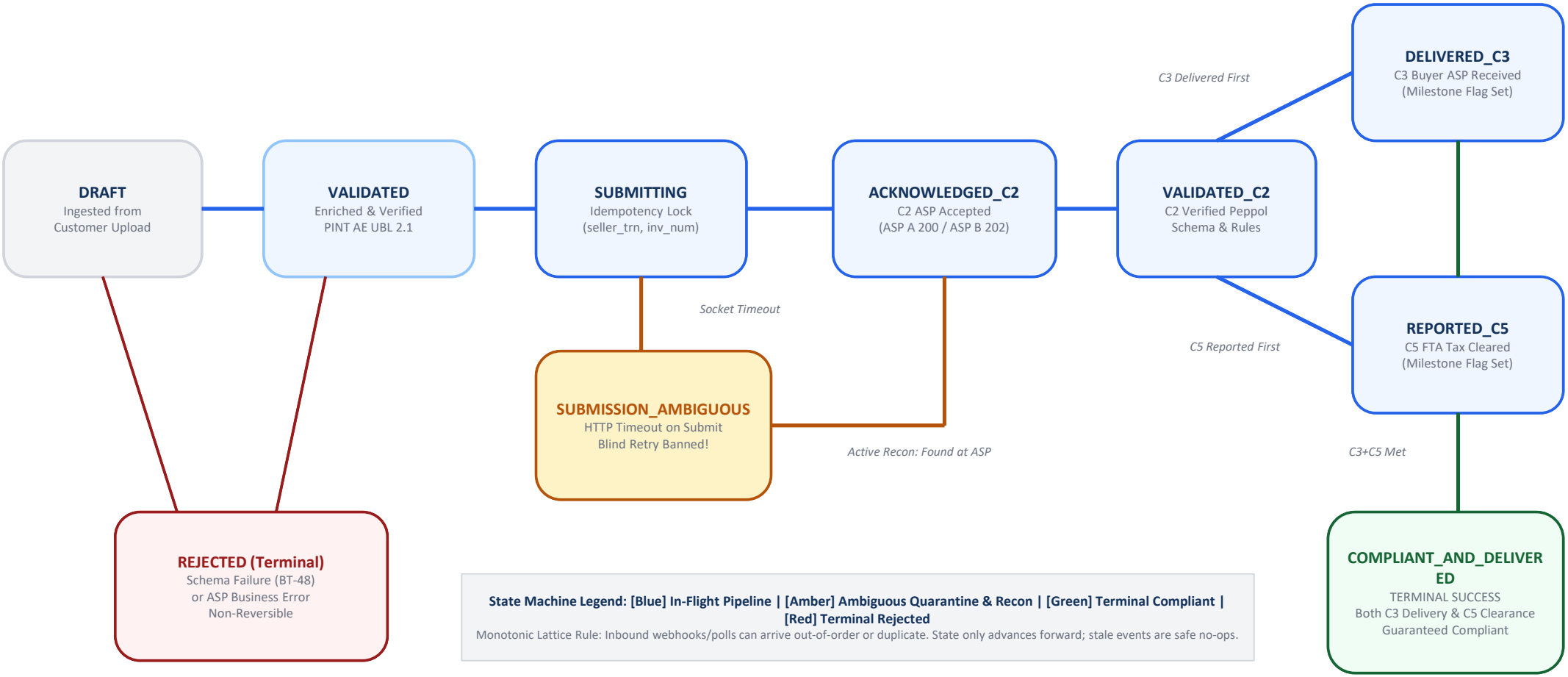
End-to-End Invoice Lifecyle: From Seller ERP (C1) through ASPs (C2 & C3) to Buyer (C4) with FTA Clearance (C5)



Sovereign Tax Data Invariant: No invoice can ever be lost, and no duplicate invoice can ever be filed with the UAE FTA.

Lifecycle State Machine Flow & Milestone Lattice

All elements below are native, fully editable PowerPoint shapes. Click, re-color, drag, or edit as needed.



State Machine Legend: [Blue] In-Flight Pipeline | [Amber] Ambiguous Quarantine & Recon | [Green] Terminal Compliant | [Red] Terminal Rejected

Monotonic Lattice Rule: Inbound webhooks/polls can arrive out-of-order or duplicate. State only advances forward; stale events are safe no-ops.